Sec: SR STAR - A & APEX, ALL IITS | DT: 07/01/2026
Time: 3 Hrs | GTM – 11 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
B	B	D	D	D	B	C	D	A	D
11	12	13	14	15	16	17	18	19	20
C	A	A	C	C	C	B	C	C	B
21	22	23	24	25					
16	6	500	2	4					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
A	B	C	A	D	D	D	A	C	B
36	37	38	39	40	41	42	43	44	45
D	A	A	D	D	C	A	B	B	C
46	47	48	49	50					
0	296	27	5	162					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
C	B	D	A	B	C	D	B	C	C
61	62	63	64	65	66	67	68	69	70
A	C	B	D	C	C	C	A	D	A
71	72	73	74	75					
45	32	8	4	167					

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HINTS
PHYSICS

1. Standard wave eqn is represented as

$$\psi\left(\vec{r},t\right)=A\sin\left(wt-\vec{k}.\vec{r}\right)$$

Where $\vec{k}$ is in direction of propagation of wave

Thus,

$$\frac{\vec{k}}{k}=\frac{\hat{x}-2\hat{y}+\sqrt{20}\hat{z}}{\sqrt{1+4+20}}$$

$$\cos\theta=\frac{\vec{k}.i}{k}=\frac{1}{5}$$

2. $$P=\frac{a-t^2}{bx}$$

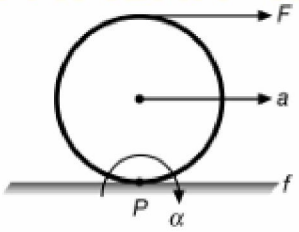
$$a=\left[t^2\right]=\left[T^2\right]$$

$$\therefore P=\frac{T^2}{bx}$$

$$b=\frac{T^2}{Px}=\frac{\left[T^2\right]}{\left[M^1L^{-1}T^{-2}\right]}=\frac{\left[T^4\right]}{\left[M^1\right]}$$

$$\frac{a}{b}=\left[T^2\right]\frac{\left[M^1\right]}{\left[T^4\right]}=\left[M^1T^{-2}\right]$$

3. Let f be the friction on the ring towards right, a be the linear acceleration and α be the angular acceleration of the ring about centre of mass.



Point of contact P is momentarily at rest, about Instantaneous Centre of mass Velocity IC (located at P), so

$$\alpha=\frac{\tau_P}{I_P}=\frac{F(2R)}{2MR^2}=\frac{F}{MR}$$

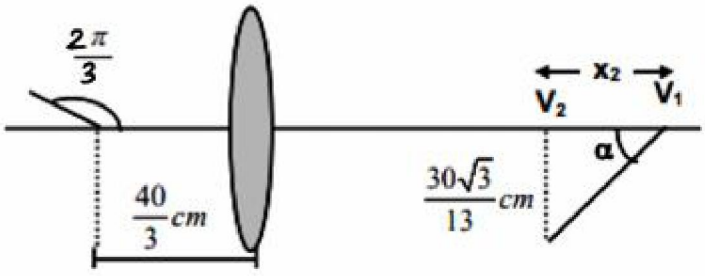
For translational motion of ring, we have

$F+f=Ma=MR\alpha=F$

$\Rightarrow f=0$

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4.



$$u_1 = -\frac{40}{3}$$

$$f = 10$$

$$\frac{1}{V_1} + \frac{1\times 3}{40} = \frac{1}{10}$$

$$V_1 = 40$$

$$u_2 = -\frac{43}{3}$$

$$\frac{1}{V_2} = \frac{1}{10} - \frac{3}{43} = \frac{43-30}{430}$$

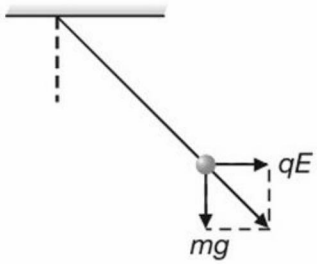
$$V_2 = \frac{430}{13}$$

$$x_2 = V_1 - V_2 = 40 - \frac{430}{13} = \frac{90}{13}$$

$$\tan\alpha = \frac{30\sqrt{3}}{13\times x} = \frac{30\sqrt{3}\times 13}{13\times 90} = \frac{1}{\sqrt{3}}$$

$$\alpha = \frac{\pi}{6}$$

5. $$a_{net} = \frac{\sqrt{m^2g^2 + q^2E^2}}{m}$$



$$\Rightarrow T = 2\pi\sqrt{\frac{l}{a_{net}}}$$

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$$\Rightarrow T = 2\pi \sqrt{\frac{l}{\sqrt{g^2 + \left(\frac{qE}{m}\right)^2}}}$$

6. 1.The swimmer must reach the opposite bank in 10 seconds. The width of the river is 40 m, so

$\upsilon_y . t = 40 \quad (1)$

Substituting $t = 10$ seconds

$\upsilon \sin\theta . 10 = 40 \Rightarrow \upsilon \sin\theta = 4 \quad (2)$

2.To reach directly across without drifting downstream, the horizontal component of the swimmer's speed must equal the speed of the river.

$\upsilon_x = \upsilon \cos\theta = 3 \quad (3)$

Now we have two equations

1. $\upsilon \sin\theta = 4$
2. $\upsilon \cos\theta = 3$

Dividing equation (2) by equation (3)

$$\frac{\upsilon \sin\theta}{\upsilon \cos\theta} = \frac{4}{3}$$

This simplifies to

$$\tan\theta = \frac{4}{3}$$

To find θ

$$\theta = \tan^{-1}\left(\frac{4}{3}\right)$$

Calculating this gives

$\theta \approx 53^\circ$

Since the angle made by the direction he swims and the river flow direction is θ, we need to find the angle α that is complementary to θ.

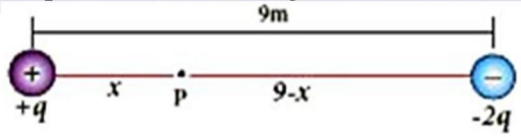
Angle with respect to river flow $= 180^\circ - \theta$

Thus

Angle $= 180^\circ - 53^\circ = 127^\circ$

7. According to the superposition principle, the total electric potential at a point is equal to the sum of the potentials due to each charge at that point.

Consider the point at which the total potential zero is located at a distance x from the charge +q as shown in the figure



The total electric potential at P is zero

$$V_{tot} = \frac{1}{4\pi \in_o}\left(\frac{q}{x} - \frac{2q}{(9-x)}\right) = 0$$

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Which gives $\frac{q}{x}=\frac{2q}{(9-x)}$

Or $\frac{1}{x}=\frac{2}{(9-x)}$

Hence, $x=3m$

8. Energy stored in powerbank = 5V×3A×2h

$=30\times3600$ J

Fraction of energy wasted is 15%

Thus, wasted energy

$=0.15\times30\times3600$ J

=16200 J

9. $A_R=\sqrt{\left(\sqrt{3}\right)^2+(1)^2}=2$

$\Rightarrow\phi=\tan^{-1}\left(\frac{\sqrt{3}}{1}\right)=\frac{\pi}{3}$

$\Rightarrow y=2\sin\left(\omega t+\frac{\pi}{3}\right)$

$\Rightarrow\frac{d^2y}{dt^2}=a=-2\omega^2\sin\left(\omega t+\frac{\pi}{3}\right)$

$\Rightarrow|a_{max}|=2\omega^2=g$

For this maximum acceleration, the mass just breaks off the plank, so $\omega=\sqrt{\frac{g}{2}}$.

This will happen for the first time when

$\omega t+\frac{\pi}{3}=\frac{\pi}{2}$

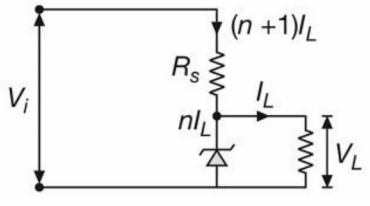
$\Rightarrow\omega t=\frac{\pi}{6}$

$t=\frac{\pi}{6\omega}=\frac{\pi}{6}\sqrt{\frac{2}{g}}=\frac{\sqrt{2}}{6}\left(\frac{\pi}{g}\right)$

10. $x_m=(\mu_r-1)$

$\Rightarrow x_m=(5500-1)=5499$

11.



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Voltage drop across Zener diode is V_L, so voltage drop across R_S is,

$$V_{RS} = V_i - V_L = (n+1) I_L R_S$$

12. Since $C_V = \dfrac{dU}{dT}$ {at constant volume}

$$C_P = \left(\frac{dQ}{dT}\right)_P \quad \{\text{at constant pressure}\}$$

$$C = \frac{dU + dW}{dT} \quad \{\text{for any process other than an adiabatic process}\}$$

$$\Rightarrow C = \frac{dU}{dT} + \frac{PdV}{dT} \quad \{\because \text{work done } dW = PdV\}$$

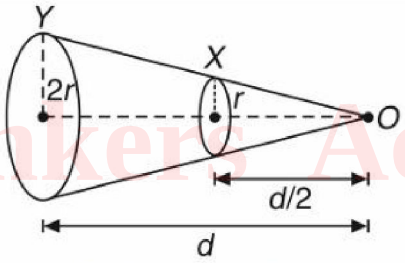
13. In the first case, $X_L = \omega L = 2\pi x 50 L = 10\pi L$

In second case, $X_L' = \omega' L = 2\pi x 100 L = 200\pi L = 2X_L$

$$i_L = \frac{100}{X_L} = I, i_L' = \frac{100}{X_L'} = \frac{100}{2X_L} = \frac{I}{2}$$

In R, current remains same.

14.



The two coils subtend the same solid angle at O and since

$$\Rightarrow \text{Solid angle} = \frac{\text{Area}}{(\perp \text{distance})^2}$$

Hence area of coil Y is 4 times the area of coil X

$\Rightarrow$ Radius of coil Y is 2 times the radius of coil X

Since the magnetic field at the axis of the coil is

$$B = \frac{\mu_0 NIR^2}{2\left(R^2 + x^2\right)^{\frac{3}{2}}}$$

$$\Rightarrow B_Y = \frac{\mu_0}{4\pi} \times \frac{2\pi I (2r)^2}{\left[(2r)^2 + (d^2)\right]^{\frac{3}{2}}}$$

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$$\Rightarrow B_X = \frac{\mu_0}{4\pi} \times \frac{2\pi I (2r)^2}{\left[r^2 + \left(\frac{d}{2}\right)^2\right]^{\frac{3}{2}}}$$

$$\Rightarrow \frac{B_Y}{B_X} = \frac{4}{\left(4r^2 + d^2\right)^{\frac{3}{2}}} \times \left[\frac{4r^2 + d^2}{4}\right]^{\frac{3}{2}}$$

$$\Rightarrow \frac{B_Y}{B_X} = \frac{4}{(4)^{\frac{3}{2}}} = \frac{4}{8} = \frac{1}{2}$$

15. $\frac{4}{5}mg - \frac{3}{5}qE = \frac{mv^2}{E} \ldots\ldots\ldots (i)$

$\frac{1}{5}mgR + \frac{3}{5}qER = \frac{1}{2}mv^2 \ldots\ldots\ldots (ii)$

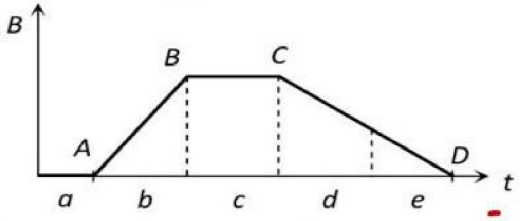
$\therefore \frac{mg}{qE} = \frac{9}{2}$

16. $I_{SYS} = \frac{ml^2}{12}\sin^2 45 + \frac{ml^2}{12}\sin^2 45$

$= \frac{ml^2}{12}$

17. Induced emf $\in = A\frac{dB}{dt}$

i.e., $\in \propto \frac{dB}{dt}$ (=slope of B – t graph)



In the given graph slope of AB>slope of CD,
Slope in the 'a' region = slope in the 'c' region = 0,
Slope in the 'd' region = slope in the 'e' region $\neq 0$
That's s why b> (d=e)>(a=c)

18. P. $\rightarrow (4)$; Q. $\rightarrow (2)$; R. $\rightarrow (1)$; S. $\rightarrow (3)$

P $KE = \frac{3}{2}K'T \Rightarrow \left[ML^2T^{-2}\right] = K'[K] \Rightarrow K' = \left[ML^2T^{-2}K^{-1}\right]$

Q. $F = 6\pi\eta rv \Rightarrow \left[ML^2T^{-2}\right] = \eta[L]\left[LT^{-1}\right] \Rightarrow \eta\left[ML^{-1}T^{-1}\right]$

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R. $E = hf \Rightarrow \left[ML^2T^{-2}\right] = \dfrac{h}{[T]} \Rightarrow h = \left[ML^2T^{-1}\right]$

S. $\dfrac{dQ}{dt} = \dfrac{K'A(\Delta T)}{\Delta x} \Rightarrow \dfrac{\left[ML^2T^{-2}\right]}{[T]} = \dfrac{k\left[L^2\right][K']}{[L]}$

$K' = \left[MLT^{-3}K^{-1}\right]$

19. For the motion of the electron along Y-axis

$$d = \frac{1}{2}\frac{eE}{m}t^2 \qquad (1)$$

Where E is the electric field due to the charged plate given by

$$E = \frac{\sigma}{\varepsilon_0} \qquad (2)$$

Where σ is the surface density of charge of the plate.

The time taken by the electron is

$$t = \frac{\ell}{u} \qquad (3)$$

From equations (1),(2) and (3), we obtain

$$d = \frac{1}{2}\frac{e\sigma}{\varepsilon_0 m}\left(\frac{\ell}{u}\right)^2$$

$$\Rightarrow \sigma = \frac{2d\varepsilon_0 mu^2}{e\ell^2}$$

20. $$E = \frac{hc}{\lambda} - \phi$$

$$2E = \frac{hc}{\lambda'} - \phi$$

$$\Rightarrow 2E - E = hc\left[\frac{1}{\lambda'} - \frac{1}{\lambda}\right] = E$$

$$\Rightarrow E + \frac{hc}{\lambda} = \frac{hc}{\lambda'}$$

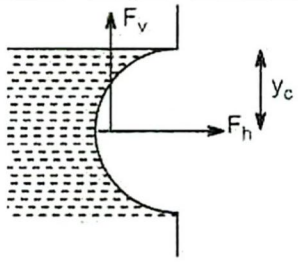
$$\Rightarrow \frac{E\lambda + hc}{\lambda} = \frac{hc}{\lambda'}$$

$$\Rightarrow \lambda' = \left(\frac{hc\lambda}{E\lambda + hc}\right)$$

21. (a) Let the horizontal and vertical thrusts on the horizontal and vertical thrusts on the hemisphere be F_h and F_v respectively

We know that

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$F_h = \rho g y_c A_v$

Where $y_c = R$

And $A_v = \pi R^2$

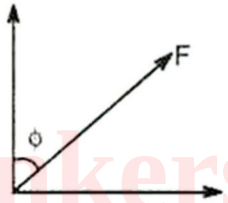
This gives $F_h = \rho g \pi R^3$ (right)

(b) Similarly using the formula $F_v = \rho V g$

Where V = Volume of the hemisphere = $\frac{2}{3}\pi R^3$,

We have $F_v = \frac{2}{3}\rho g \pi R^3 \quad (\text{up})$

(c) Hence the net hydrostatic force on the hemisphere is

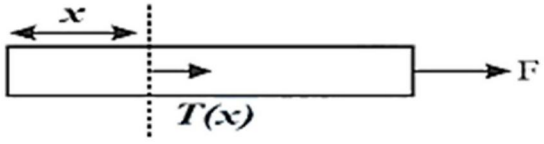


$F = \sqrt{F_h^2 + F_v^2}$

$F = \sqrt{\left(\rho g \pi R^3\right) + \left(\frac{2}{3}\rho g \pi R^3\right)^2}$

$= \frac{\sqrt{13}}{3}\rho g \pi R^3$

22. $U = \frac{F^2 l}{6A^2 Y}$



The acceleration of the rod is

$a = \frac{F}{m}$

So, the tension in the rod at a distance x from the free end is

$T(x) = \frac{mx}{l} \times \frac{F}{m} = \frac{Fx}{l}$

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The longitudinal stress developed inside the rod at a distance x from the free end is

$$\sigma = \frac{T(x)}{A} = \frac{Fx}{lA}$$

The strain energy density (energy per unit volume) is

$\frac{dU}{dV} = \frac{1}{2}\frac{\sigma^2}{Y} = \frac{1}{2}\frac{F^2x^2}{YA^2l^2}$, Adx is the volume of a small part of the rod

So the energy stored in the rod is

$$U = \frac{1}{2}\frac{F^2}{YA^2l^2}\int_0^l x^2dx$$

$$U = \frac{F^2l}{6A^2Y}$$

23. $a\sin\theta = 1\times\lambda$

$$\Rightarrow \lambda = a\sin\theta = 10^{-4}\times\frac{1}{2} = \frac{10\times10^{-5}}{2}$$

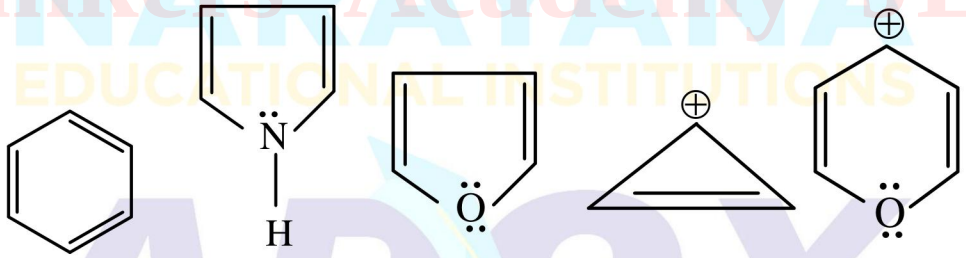
$$= 5\times10^{-5} \text{ cm} = 5000\text{A}$$

24. Each vibrational mode is associated with two degrees of freedom, thus 6 vibrational Do F divided by 3 translational DoF

25. $\text{avg.vel} = \frac{2v_1v_2}{v_1+v_2}$

CHEMISTRY

26.



(a) (d) (e) (f) (g)

a , d , e , f, g Follow Hukels Rule

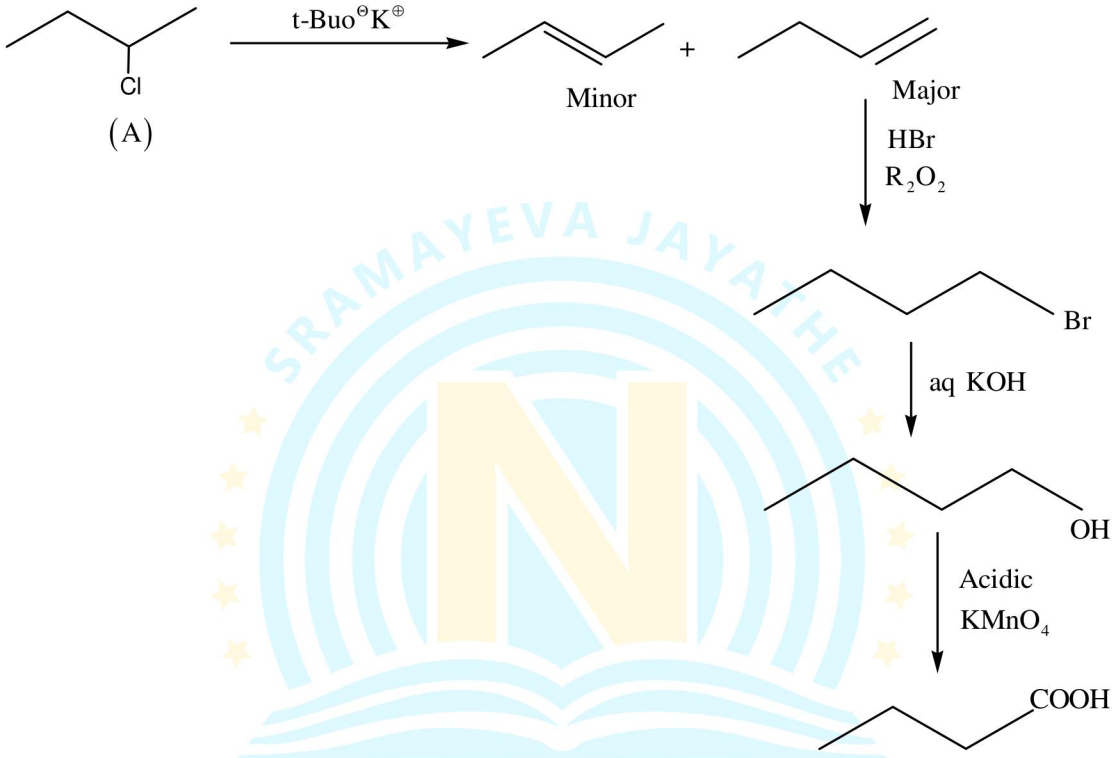


(b) (c)

b , c not Aromatic do not follow Hukels Rule

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27.



28. Conceptual

29. partial pressure $\propto$ mole fraction of gas

30. $r \propto n^2$

$$\frac{r^5}{r_1} = \frac{25}{1} = 25$$

$$\frac{r_6}{r_2} = \frac{36}{4} = 9$$

$$\frac{r_4}{r_2} = \frac{16}{4} = 4$$

$$\frac{r_2}{r_1} = \frac{4}{1} = 4$$

31. $H_2O_{(l)} \rightarrow H_2O_{(g)};\ \Delta H = \Delta E + \Delta nRT$

32. $P_T = P_A^o \times A + P_B^o \times B$

$$500 = 100 \times \frac{1}{4} + P_B^o \times \frac{3}{4}$$

$$500 = 25 + P_B^o \times \frac{3}{4}$$

$$475 = P_B^o \times \frac{3}{4}$$

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$P_B^o = \frac{1\times 475}{3}$

$=633\,Hg$

$P_A^o < P_B^o$ A is least volatile

33. $CaCO_3$ molar mass = 100 g/mol

$CaCl_2$ molar mass = 111 g/mol

$\eta_{CaCO_3} = \frac{500}{100} = 5$ mol

$\eta_{HCl} = 1.20\times 0.3L = 0.360$ mol

$CaCO_3(s) + 2HCl \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$

5mole 0.360 mol

↓

L.R

Moles of $CaCl_2 = \frac{0.360}{2} = 0.180$ mol

$mass\ of\ CaCl_2 = 0.180\times 111$

$= 19.98g$

34. $A_2B_3 \rightleftharpoons (2S)^2 \times (3S)^3 ; Ksp = 108S^5$

35.

Molecule	Hybridization	D.M	L.P on C.A
ClO_2	SP^2	Non-zero	1
PCl_3	SP^3	Non-zero	1
BrF_3	SP^3d	Non-zero	2
SF_6	SP^3d^2	Zero	0
xeF_2	SP^3d	Zero	3

36. OH group increases reactivity, soluble in base, salicyaldehyde has no $\alpha - H$ so no aldol

37. Leucine & Threonine have 2 chiral centres
Aspartic acid has –COOH in side chain, not – OH
Sulphur only in methionine and cysteine

38. Basic strength of Aliphatic amine based on Hydrogen bonding and electronic inductive effect.

39. Metallic radius order (Group 13)
B < Al < Ga < In < Tl
So In > Ga Statement I correct

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Ionic radius order $(M+3)$

$B^{+3} < Al^{+3} < Ga^{+3} < In^{+3} < T_l^{+3}$

So $Ga^{+3} > Al^{+3}$ Statement II correct

40. S.F.L $\rightarrow \Delta_o \uparrow \rightarrow \Delta_o > P \rightarrow \varepsilon^{\odot}$'s

Pair $\rightarrow$ lowspin

Tetrahedral splitting energy $\Delta_t = \frac{4}{9}\Delta_0$

Always small $\rightarrow \Delta_t < P \rightarrow \rightarrow$ no pairing

High spin always

41. Conceptual

42. Moles of CO_2

$n = \frac{2.20}{44} = 0.050$

Moles of C = 0.050

Mass of C = $0.050 \times 12 = 0.60$ g

Moles of H_2O

$n = \frac{0.900}{18} = 0.050$

Moles of H = $2 \times 0.050 = 0.10$ mol

Mass of H = $0.10 \times 1 = 0.10$ g

Mass of O in compound

Total mass = 1.50 g

$W_O = 1.50 - (0.60 + 0.10) = 0.80$g

Moles of O:

$\frac{0.80}{16} = 0.05$

Mole ratio

C:H:O

0.050 : 0.10 : 0.05

Divide all by 0.050

C:H:O = 1:2:1

Empirical formula = CH_2O

Empirical formula mass = 12+2+16 = 30

43. II is more stable because resonance

I is least stable because alkenyl carbanion

44. Rate of Hydrolysis α leaving group ability

Leaving group order : $Cl^- > R-COO^- > Ro^{\Theta} > -NH-R$

Acidchloride > Anhydride > ester > Amide

45. Rare, monoatomic ,stable p-block element – Xenon (Xe)

Highest electronegativity – Fluorie (F)

Second highest – Oxygen (O)

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So molecule = XeF_4O_2 or equivalent

Central atom A = Xe has 6 electron pairs around it $\left(AX_4Y_2\right)$

4 bonding pairs with F

2 bonding pairs with O

No lone pair described (since 6 ligands already use all 6 positions of octahedral geometry)

6 bonds – ideal geometry = octahedral

46. Among V, Cr, Mn, Fe the highest standard potential for the couple M^{3+}/M^{2+} is for Mn (so M = Mn)

In $\left[M\left(CN\right)_6\right]^{3-}$ each $CN^- = -1$, total ligand charge = -6. Complex charge is -3, so metal oxidation state x satisfies

$x+\left(-6\right)=-3 \Rightarrow x=+3$. Thus we have Mn (III)

Mn (III) is d^4. CN^- is a strong field ligand so the complex is low-spin (octahedral)

Electrons fill t_{2g} first $\rightarrow t_{2g}^4 e_g^0$

Therefore the number of electrons in the e_g orbital = 0

47. $$E_{cell} = 3.17 - \frac{0.06}{2} \times log \frac{0.1}{\left(0.0001\right)^2}$$

48. Molecular weight of $x = 155$, molecular weight of $y = 115$

Molar Mass = 372 gm

$\therefore 0.2$ mole has $0.2 \times 372 = 74.4$ g

49. The reaction for the decomposition of methanol is:

$$CH_3OH\left(g\right) \rightleftharpoons CO\left(g\right) + 2H_2\left(g\right)$$

The equilibrium constant expression for K_p is:

$$K_p = \frac{P_{CO} \cdot \left(P_{H_2}\right)^2}{P_{CH_3OH}}$$

The given equilibrium partial pressures are

$P_{CH_3OH} = 2.0atm$, $P_{CO} = 1.0atm$, and

$P_{H_2} = 0.1atm$. Substituting these values into the

K_p expression:

$$K_p = \frac{\left(1.0\right).\left(0.1\right)^2}{2.0}$$

$$K_p = \frac{1.0 \cdot 0.01}{2.0}$$

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$$K_p = \frac{0.01}{2.0}$$

$$K_p = 0.005 atm^2$$

$$K_p = 5 \times 10^{-3} atm^2$$

50. Since graph of $t_{1/2}$ vs $[A]_0$ is linear

$t_{1/2} \propto [A]_0 \Rightarrow$ Zero order reaction

Half-life for zero order

$$t_{1/2} = \frac{[A]_0}{2k}$$

Given slope = 54.0

$$\text{Slope} = \frac{1}{2k} = 54$$

Solve for k

$$k = \frac{1}{2 \times 54} = \frac{1}{108} = 9.259 \times 10^{-3} \text{ mol L}^{-1} \text{ min}^{-1}$$

$$t_{1/2} = \frac{[A]_0}{2K}$$

$$= \frac{3.0}{2\left(9.26 \times 10^{-3}\right)}$$

$$t_{1/2} = \frac{3.0}{0.01852}$$

$$t_{1/2} = 162 \text{ min}$$

MATHEMATICS

51. $$\text{Exponents of } 5 = \left\lfloor \frac{100}{5} \right\rfloor + \left\lfloor \frac{100}{5^2} \right\rfloor + \left\lfloor \frac{100}{5^3} \right\rfloor$$

$$\text{First term : } \left\lfloor \frac{100}{5} \right\rfloor = \lfloor 20 \rfloor = 20$$

$$\text{Second term : } \left\lfloor \frac{100}{25} \right\rfloor = \lfloor 4 \rfloor = 4$$

$$\text{Third term : } \left\lfloor \frac{100}{125} \right\rfloor = \lfloor 0.8 \rfloor = 0$$

Adding these together

Exponents of 5 = 20 + 4 + 0 = 24

$$\text{Exponent of } 2 = \left\lfloor \frac{100}{2} \right\rfloor + \left\lfloor \frac{100}{4} \right\rfloor + \left\lfloor \frac{100}{8} \right\rfloor$$

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$$+\left\lfloor\frac{100}{16}\right\rfloor+\left\lfloor\frac{100}{32}\right\rfloor+\left\lfloor\frac{100}{64}\right\rfloor$$

First term : $\lfloor 50\rfloor = 50$

Second term : $\lfloor 25\rfloor = 25$

Third term : $\lfloor 12.5\rfloor = 12$

Fourth term : $\lfloor 6.25\rfloor = 6$

Fifth term : $\lfloor 3.125\rfloor = 3$

Sixth term : $\lfloor 1.5625\rfloor = 1$

Adding these together

Exponent of 2 = 50 + 25 + 12 + 6 + 3 +1 = 97

Number of trailing zeros = min(Exponent of 2,Exponent of 5) = min(97,24) = 24 = 24

The number of zeros in the expansion of 100! is 24

52. Let the semi-major axis of the ellipse be a_1 and the semi-minor axis be b_1.The eccentricity e_1 of the ellipse is given by

$$e_1 = \sqrt{1-\frac{b_1^2}{a_1^2}}$$

For the hyperbola, let the semi-major axis be a_2 and the semi-minor axis be b_2. The eccentricity e_2 of the hyperbola is given by

$$e_2 = \sqrt{1+\frac{b_2^2}{a_2^2}}$$

We know that the conjugate axis of the hyperbola is equal to the mirror axis of the ellipse

$b_2 = b_1 = b$

Since both conics are confocal , we have the relationship between the axes

$a_1 e_1 = a_2$

This implies $a_1 = \frac{a_2}{e_1}$

Substitute b_1 and b_2

`Now substituting $b_1 = b_2 = b$ into the eccentricity formulas

$$e_1^2 = 1-\frac{b^2}{a_1^2}$$

$$e_2^2 = 1+\frac{b^2}{a_2^2}$$

Express $\frac{1}{e_1^2}$ and $\frac{1}{e_2^2}$

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From the eccentricity formula, we can express

$$\frac{1}{e_1^2}=\frac{1}{1-\frac{b^2}{a_1^2}}=\frac{a_1^2}{a_1^2-b^2}$$

$$\frac{1}{e_2^2}=\frac{1}{1-\frac{b^2}{a_2^2}}=\frac{a_2^2}{a_2^2-b^2}$$

Now , we need to add these two fraction

$$\frac{1}{e_1^2}+\frac{1}{e_2^2}=\frac{a_1^2}{a_1^2-b^2}+\frac{a_2^2}{a_2^2+b^2}$$

$$=\frac{a_1^2\left(a_2^2+b^2\right)+a_2^2\left(a_1^2-b^2\right)}{\left(a_1^2-b^2\right)\left(a_2^2+b^2\right)}$$

Expanding the numerator

$$=\frac{a_1^2a_2^2+a_1^2b^2+a_2^2a_1^2-a_2^2b^2}{\left(a_1^2-b^2\right)\left(a_2^2+b^2\right)}$$

$$=\frac{2a_1^2a_2^2+\left(b^2\right)\left(a_1^2-a_2^2\right)}{\left(a_1^2-b^2\right)\left(a_2^2+b^{2a}\right)}$$

Use the relationship $a_1^2-a_2^2=2b^2$

From the previous steps, we have

$a_1^2-a_2^2=2b^2$

Substituting this into our expression gives

$$=\frac{2a_1^2a_2^2+2b^2b^2}{\left(a_1^2-b^2\right)\left(a_2^2+b^2\right)}$$

Final simplification

After simplification, we find that

$$\frac{1}{e_1^2}+\frac{1}{e_2^2}=2$$

53. Let x = 2 y + z = 13

$(4,9)$, $(5,\vec{8})$, $(6,7)$, $(7,6)$, $(8,5)$,$(9,4)$,6

Let x = 3 y + z = 12

$(3,9),(4,\vec{8}),\ldots\ldots,(9,3)$ 7

Let x = 4 y + z = 11

$(2,9),(3,\vec{8}),\ldots\ldots,(9,1)$ 9

Let x = 5 y + z = 10

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$(1,9),(3,8\vec{})$,.....$(9,1)$ 10

Let x = 6 y + z = 9

$(0,9),(1,8\vec{})$,.....,$(9,0)$ 9

Let x = 7 y + z = 8

$(0,9),(1,7\vec{})$,.....,$(8,0)$ 9

Let x = 8 y + z = 7

$(0,7),(1,6\vec{})$,.....,$(7,0)$ 8

Let x = 9 y + z = 6

$(0,6),(1,5\vec{})$,.....,$(6,0)$ 7

Total = 6 + 7 + 8 + 9 + 10 + 9 + 8 + 7 = 64

54. $f\left(\frac{x+2y}{3}\right)=\frac{f(x)+2f(y)}{3}$

We also know that $f'(0)=1$ and $f(0)=2$. We need to find the function $f(x)$. The functional equation resembles the section formula, which suggest that $f(x)$ could be a linear function. We assume a linear form for $f(x)$

$f(x)=ax+b$

Substituting $f(x)=ax+b$ into the functional equation

$$f\left(\frac{x+2y}{3}\right)=a\left(\frac{x+2y}{3}\right)+b=\frac{ax+2ay}{3}+b$$

$$\frac{f(x)+2f(y)}{3}=\frac{(ax+b)+2(ay+b)}{3}$$

$$=\frac{ax+b+2ay+2b}{3}=\frac{ax+2ay+3b}{3}$$

Now we set both sides of the equation equal to each other

$$\frac{ax+2ay}{3}+b=\frac{ax+2ay+3b}{3}$$

Multiplying through by 3 to eliminate the denominator gives

$ax+2ay+3b=ax+2ay+3b$

This equation holds true for all x and y, confirming that our assumption of $f(x)$ being linear is valid

Using the conditions $f'(0)=1$ and $f(0)=2$.

From the linear function $f(x)=ax+b$:

$f(0)=b=2$

The derivative $f'(x)=a$. Given $f'(0)=1$

$a=1$

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Now substituting a and b back into the linear function

$f(x)=1\ x+2=x+2$

Conclusion

Thus, the required function is

$f(x)=x+2$

55. $\left|\text{adj}\left(A-2A^{T}\right)\left(2A-A^{T}\right)\right|=28$

$\left|\left(A-2A^{T}\right)\left(2A-A^{T}\right)\right|=24$

$\left|A-2A^{T}\right|\left|2A-A^{T}\right|=\pm 16$

$\left|A-2A^{T}\right|=A^{T}-2A$

$\left|A-2A^{T}\right|$

$=\left|A^{T}-2A\right|$

$\Rightarrow\left|A-2A^{T}\right|^{2}=16$

$\left|A-2A^{T}\right|=\pm 4$

$\begin{bmatrix}1&2&\alpha\\1&0&1\\0&1&2\end{bmatrix}$

$-\begin{bmatrix}2&2&0\\4&0&2\\2\alpha&2&4\end{bmatrix}$

$\begin{vmatrix}-1&0&\alpha\\-3&0&-1\\-2\alpha&-1&-2\end{vmatrix}$

$1+3\alpha=4$

$3\alpha=3$

$\alpha=1$

$|A|=\begin{vmatrix}1&2&1\\1&0&1\\0&1&2\end{vmatrix}=-1-3=-4$

$|A|^{2}=16$

56. $\left(1-x^{2}\right)\left(x+2/x\right)^{6}=\left(x+2/x\right)^{6}-x^{2}\left(x+2/x\right)^{6}$

Consider $\left(x+2/x\right)^{6}$

Here, $a=x,\ b=2/x,\ n=6$

We have

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$t_{r+1} = {}^{n}C_r a^{n-r}.b^r$

$= {}^{6}C_r x^{6-r}\left(\frac{2}{x}\right)^r$

$= {}^{6}C_r x^{6-r} 2^r . x^{-r}$

$= {}^{6}C_r x^{6-2r} 2^r$

To get the term independent of x in the expansion of $(x+2/x)^6$, we must have

$x^{6-2r} = x^0$

$\therefore 6-2r=0$

$\therefore r=3$

To get the term independent of x in the expansion of $x^2(x+2/x)^6$, we must have

$x^2 \quad x^{6-2r} = x^0$

$\therefore 8-2r=0$

$\therefore r=4$

$\therefore$ Term independent of x in the expansion of

$\left(1-x^2\right)\left(x+\frac{2}{x}\right)^6 = {}^{6}C_r 2^r - {}^{6}C_r 2^r$

$= {}^{6}C_3 2^3 - {}^{6}C_4 2^4$

$= 20(8) - 15(16)$

$= 160 - 240$

$= -80$

57. $2\cos\theta + 2\sqrt{2} = 3\sec\theta$

$2\cos^2\theta + 2\sqrt{2}\cos\theta - 3 = 0$

$\cos\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta \in Q_1 \text{ or } Q_4$

$\Rightarrow \sin\theta = -\frac{1}{\sqrt{2}}, \cot\theta = -1, \tan\theta = 1$

58. Correct option is b.2^{10}

Put x = i and x = -i

$\Rightarrow (1+i)^{10} = (a_0 - a_2 + a_4 - a_6 + a_8 - a_{10}) + i(a_1 - a_3 + a_5 + a_5 - a_7 + a_9) \ldots (1)$

Also,

$\Rightarrow (1-i)^{10} = (a_0 - a_2 + a_4 - a_6 + a_8 - a_{10}) - i(a_1 - a_3 + a_5 + a_5 - a_7 + a_9) \ldots (2)$

Multiply (1) and (2)

$((1+i)(1-i)^{10} = (a_0 - a_2 + a_4 - a_6 + a_8 - a_{10})^2 + (a_1 - a_3 + a_5 - a_7 + a_9)^2$

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$$2^{10}=\left(a_0-a_2+a_4-a_6+a_8-a_{10}\right)^2+\left(a_1-a_3+a_5-a_7+a_9\right)^2$$

59. $p(x)=2x^3-3ax^2-12a^2x-2$

$p'(x)=6x^2-6ax-12a^2$

$p'(x)=0\ given$

$x^2-ax-2a^2=0$

$$x=\frac{a\pm\sqrt{a^2+8a^2}}{2}=\frac{a+3a}{2}=2a\ or-a$$

$p''(x)=12x-6a$

$p''(-a)=-12a-6a=-18a$

$p''(2a)=24a-6a=18a$

$$p\left(\frac{s+t}{2}\right)=p\left(\frac{2a+(-a)}{2}\right)=p\left(\frac{a}{2}\right)=-54\,(\text{given})$$

$$\Rightarrow\frac{a^2}{4}-\frac{3a^3}{4}-\frac{12a^3}{2}-2=-54$$

$\Rightarrow -2a^3-24a^3=-208$

$\Rightarrow a^3=-208/-26=8=2^3$

$\Rightarrow a=2$

$\therefore f'(-a)=-18\times2<0$

$f''(2a)=18\times2>0$

$\therefore x=-a$ is point of maxima

$x=2a$ is point of minima

60. $$\text{Re}\left(\frac{z-1}{2z+i}\right)+\text{Re}\left(\frac{\bar{z}-1}{2\bar{z}-i}\right)=2$$

$$\frac{z-1}{2z+i}=\left(\frac{\overline{\bar{z}-1}}{2\bar{z}-i}\right)=2$$

Here,

$$\text{Re}\left(\frac{z-1}{2z+i}\right)+\text{Re}\left(\overline{\frac{z-1}{2z+i}}\right)=2$$

$$=2\text{Re}\left(\frac{z-1}{2z+i}\right)=2$$

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$$\Rightarrow \text{Re}\left(\frac{z-1}{2z+i}\right)=1$$

Let z = z + iy

$$\text{Re}\left(\frac{(x-1)+iy}{2x+i(2y+1)}\right)=1$$

$$\Rightarrow \text{Re}\left[\frac{((x-1)+iy)(2x-i)(y+1)}{(2x+i(2y+1)(2x-i(2y+1))}\right]=1$$

$$\Rightarrow \frac{2x(x-1)+y(2y+1)}{4x^2+(2y+1)^2}=1$$

$$\Rightarrow 2x^2-2x+2y^2+y=4x^2+4y^2+1+4y$$

$$\Rightarrow 2x^2+2y^2+3y+2x+1=0$$

$$\Rightarrow x^2+y^2+x+\frac{3}{2}y+\frac{1}{2}=0$$

centre

$$=\left(\frac{-1}{2},\frac{-3}{4}\right),r$$

$$=\sqrt{\frac{1}{4}+\frac{9}{16}-\frac{1}{2}}$$

$$=\frac{\sqrt{5}}{4}$$

$$a=\frac{-1}{2},b=\frac{-3}{4}.r^2$$

$$15\frac{ab}{r^2}=15\times\left(\frac{-1}{2}\right)\times\left(\frac{-3}{4}\right)\times\frac{16}{5}=18$$

61. $\vec{b}\times\vec{d}=\vec{c}\times\vec{d}$ and $\vec{a}\,.\,\vec{b}=4$

$$\Rightarrow \vec{d}=\lambda\left(\vec{b}-\vec{c}\right)=\lambda\left(\hat{i}-2\hat{j}+\hat{k}\right)$$

$\because \vec{a}\,.\,\vec{b}=4\Rightarrow\lambda=-2$

Also,

$$\left|\vec{a}\times\vec{d}\right|^2+\left|\vec{a}\,.\,\vec{d}\right|^2=\left|\vec{a}\right|^2\left|\vec{d}\right|^2$$

$$\Rightarrow\left|\vec{a}\times\vec{d}\right|^2=6\times4\times6-16=128$$

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62. Given relation is $R=\{(a,b):1+ab>0\}$

Reflexive A relation $R\subseteq A\times A$ is reflexive, if $(a,a)\in R$

Now, $R=\{(a,b)1+ab>0\}on Z.$

$\Rightarrow (a,a)\in R\Rightarrow 1+a^2>0$

So, it is reflexive.

Symmetric if $(a,a)\in R\Rightarrow (b,a)\in R$

Since, $1+ab>0\Rightarrow 1+ba>0$

$\Rightarrow (b,a)\in R$

So, it is symmetric.

Transitive if $(a,b)\in R and (b,c)\in R\Rightarrow (a,c)\in R$

Now, $(a,b)\in R and (b,c)\in R$

$\Rightarrow 1+ab>0 and 1+bc>0$

$\Rightarrow 2+b(a+c)>0$

$i.e.,(a,c)\notin R$

So, it is not transitive.

63. At $x\to 0$

$\sin 2x-\beta x\to 0$

$\Rightarrow \frac{0}{0} form$

$\Rightarrow (\gamma-1)e^0+\sin(\alpha x)\to 0$

$\Rightarrow (\gamma-1)=0$

$\Rightarrow \gamma=1$

$$\Rightarrow \lim_{x\to 0}\frac{x^2\sin(\alpha x)}{(\sin 2x-\beta x)}=3$$

$$\Rightarrow \lim_{x\to 0}\frac{x^2\left[\alpha x-\frac{(\alpha x)^3}{3!}+\frac{(\alpha x)^5}{5!}-\ldots\right]}{(2x)-\left[\frac{(2x)^3}{3!}+\frac{(2x)^5}{5!}\right]-\beta x}$$

$$\Rightarrow \lim_{x\to 0}\frac{\alpha x^3-\frac{\alpha^3x^5}{3!}+\frac{\alpha^5x^7}{5!}-\ldots.}{x(2-\beta)-\frac{8x^3}{6}+\frac{2^5.x^5}{5!}-\ldots}=3$$

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$\Rightarrow 2-\beta=0 and \frac{\alpha}{\frac{-8}{6}}=3$

$\Rightarrow \beta=2$

$\alpha=3\left(-\frac{8}{6}\right)=-4$

$\Rightarrow \gamma=1, \beta=2, \alpha=-4$

$\Rightarrow \alpha+2\beta+\gamma=1+4-4=1$

64. $D=\begin{vmatrix}1 & -2 & 1\\ 2 & \alpha & 3\\ 3 & -1 & \beta\end{vmatrix}=1(\alpha\beta+3)+2(2\beta-9)+1(-2-3\alpha)$

$=\alpha\beta+3+4\beta-18-2-3$

$D=0, D_1=0$

$D_2=0_{and} D_3=0$

$D=0$

$\alpha\beta-3\alpha+4\beta=17$

$D_1=\begin{vmatrix}-4 & -2 & 1\\ 5 & \alpha & 2\\ 2 & -1 & \beta\end{vmatrix}=0$

$D_2=\begin{vmatrix}1 & -4 & 1\\ 2 & 5 & 3\\ 3 & 3 & \beta\end{vmatrix}=0$

$\Rightarrow 1(5\beta-9)+4(2\beta-9)+1(6-15)=0$

$13\beta-9-36-9=0$

$13\beta=54, \beta=\frac{54}{13}$

Put in (1)

$\frac{54}{13}\alpha-2\alpha+4\left(\frac{54}{13}\right)=17$

$54\alpha-39\alpha+216=221$

$15\alpha=5$

$\alpha=\frac{1}{3}$

Now,

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$$12\alpha+13\beta=12.\frac{1}{3}+13.\frac{54}{13}=4+54=58$$

65. To solved the problem, we are given the following information:

1. $P_n=\alpha^n+\beta^n$
2. $\alpha+\beta=1$
3. $\alpha\beta=-1$
4. $P_{n-1}=11$
5. $P_{n+1}=29$

We need to find P_n^2.

Establish the recurrence relation for P_n

Using the properties of roots, we can derive a recurrence relation for P_n:

$$P_n=(\alpha+\beta)P_{n-1}-\alpha\beta P_{n-2}$$

Substituting the known values:

$$P_n=1.P_{n-1}-(-1).P_{n-2}$$

This simplifies to:

$$P_n=P_{n-1}+P_{n-2}$$

use the given values to find P_{n-2}

We know:

$$P_{n-2}=11 \text{ and } P_{n+1}=29$$

Using the recurrence relation:

$$P_{n+1}=P_n+P_{n-1}$$

So,

$$29=P_n+11$$

This gives us:

$$P_n=29-11=18$$

Calculate P_n^2

Now that we have $P_n=18$, we can find P_n^2

$$P_n^2=18^2=324$$

$$P_n^2=324$$

66. Using the Formula for Tangents:

The formulas for $\tan(\alpha/2)$ and $\tan(\beta/2)$ in terms of the semi-perimeter s and the lengths of the sides of triangle PSS' can expressed as:

$$\tan\left(\frac{\alpha}{2}\right)=\frac{s-b}{s-a} \text{ and } \tan\left(\frac{\beta}{2}\right)=\frac{s-a}{s-c}$$

Where s is the semi-perimeter of triangle PSS'.

Calculating the semi-permeter:

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The semi-perimeter s is given by:

$$s = \frac{SP + S'P + SS'}{2} = \frac{2\alpha + SS'}{2}$$

The distance SS' can be calculated as $SS' = 2\alpha e$

Calculating $\tan(\alpha/2)$ and $\tan(\beta/2)$

Substitute s into the tangent formulas:

$$\tan\left(\frac{\alpha}{2}\right) = \frac{a(1+e)-b}{a(1+e)-a} = \frac{a(1+e)-b}{ae}$$

$$\tan\left(\frac{\beta}{2}\right) = \frac{a(1+e)-a}{a(1+e)-c} = \frac{ae}{a(1+e)-c}$$

Finding the Ratio

Now, we find the ratio:

$$\frac{\tan(\alpha/2)}{\tan(\beta/2)} = \frac{\frac{a(1+e)-b}{ae}}{\frac{ae}{a(1+e)-c}}$$

$$= \frac{(a(1+e)-b)(a(1+e)-c}{a^2e}$$

Final simplification

After simplifying, we arrive at:

$$\frac{\tan(\alpha/2)}{\tan(\beta/2)} = \frac{1-e}{1+e}$$

Thus, the value of $\frac{\tan(\alpha/2)}{\tan(\beta/2)}$ is

$$\frac{1-e}{1+e}$$

67. P (5,1,-3)

$$L_1 : x-1 = y-2 = z = \lambda$$

$$L_2 : x-2 = y = z-1 = \mu$$

Any point of L_1 is

$$Q(\lambda+1, \lambda+2, \lambda)$$

Any point of

$$L_2 \text{ is } R(\mu+2, \mu, \mu+1)$$

$$PQ < \lambda-4, \lambda+1, \lambda+3\rangle . \langle 1,1,1\rangle = 0$$

$$\lambda-4+\lambda+1+\lambda+3=0$$

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$3\lambda = 0$

$\Rightarrow \lambda = 0$

$\therefore Q(1,2,0)$

Also, $PR < \mu-3, \mu-1, \mu+4 > . < 1,1,1 = 0$

$\mu-3+\mu-1+\mu+4=0$

$\Rightarrow \mu = 0$

$R(2,0,1)$

$$\text{Area} = \frac{1}{2}\left|\overrightarrow{PQ} \times \overrightarrow{PR}\right| = \frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -1 & -3 \\ 3 & 1 & -4 \end{vmatrix}$$

$$= \frac{1}{2}\left|7\hat{i}+7\hat{j}+7\hat{k}\right| = \frac{1}{2}\times 7\times\sqrt{3} = \frac{7\sqrt{3}}{2}$$

68. Domain $x \in \left[\frac{4}{3}, \frac{25}{4}\right)$

$a_1 = 2, a_2 = 3, a_3 = 4, a_4 = 5, a_5 = 6$

So, $L: \frac{x-1}{2} = \frac{y+2}{3} = \frac{z-4}{6}$

$$A: \frac{x-1}{2} = \frac{y+2}{3} = \frac{0-4}{6} \rightarrow A\left(\frac{-1}{3}, -4, 0\right)$$

For $B: \frac{0-1}{2} = \frac{y+2}{3} = \frac{z-4}{6} \rightarrow B\left(0, \frac{-7}{2}, 1\right)$

$$C: \frac{x-1}{2} = \frac{0+2}{3} = \frac{z-4}{6} \rightarrow C\left(\frac{7}{3}, 0, 8\right)$$

So, $C^1: \left(\frac{7}{3}, 0, -8\right)$

The volume of tetrahedran $OABC^1 = \frac{1}{6}\left|\left[\overline{OA}\,\overline{OB}\,\overline{OC^1}\right]\right|$

$V = \frac{28}{9}$ cubic units

Where $\left[\overline{OA}\,\overline{OB}\,\overline{OC^1}\right] = \begin{vmatrix} -1/3 & -4 & 0 \\ 0 & -7/2 & 1 \\ 7/3 & 0 & -8 \end{vmatrix}$

i.e $90V = 280$

69. $\left|adj\left(-4adj\left(-3adj\left(2A^{-1}\right)\right)\right)\right|$

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$$= \left|-4adj\left(-3adj\left(2A^{-1}\right)\right)\right|^2$$

$$= 4^6\left|-3adj\left(2A^{-1}\right)\right|^4$$

$$= 4^6.3^{12}\left|adj\left(2A^{-1}\right)\right|^4$$

$$= 4^6.3^{12}\left|2A^{-1}\right|^8$$

$$= 4^6.3^{12}.2^{24}\left|A^{-1}\right|^8$$

$$= 4^6.3^{12}.2^{24}.\frac{1}{|A|^8}$$

$$= 3^{12}.\frac{2^{36}}{3^8}$$

$$= 3^4.2^{36}$$

$m = 36$

$n = 4$

$\Rightarrow m + 2n = 36 + 8 = 44$

70. The required parabola is $\left(y - \frac{12}{5}\right)^2 = \frac{8}{5}\left(x - \frac{3}{5}\right)$

Vertex $= \left(\frac{3}{5}, \frac{12}{5}\right)$

$a = \frac{2}{5}$

Focus $= \left(1, \frac{12}{5}\right) = (x_1, y_1)$

$: x_1 + y_1 = 1 + \frac{12}{5} = \frac{17}{5}$

71. The function (x) is constant on each interval

$(n, n+1)$ for $n = 0, 1, 2, \ldots\ldots 9$. Therefore, we can **write**:

$$I = \sum_{n=0}^{9} \int_n^{n+1} \frac{ne^n}{e^{(x-1)}}\,dx$$

The integrand simplifies to:

$$\frac{ne^n}{e^{(x-1)}} = ne^n e^{1-x} = ne^{n+1}e^{-x}$$

Thus, We have:

$$I = \sum_{n=0}^{9} ne^{n+1} \int_n^{n+1} e^{-x}\,dx$$

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The integral $\int e^{-x}dx$ can be computed as

Follows:

$\int e^{-x}dx = -e^{-x} + c$

Evaluating from n to $n+1$:

$\int_n^{n+1} e^{-x}dx = (-e^{-x})_n^{(n+1)} = -e^{(n+1)} + e^{-n} = e^{-n}$

$-e^{-(n+1)} = e^{-n}\left(1-e^{-1}\right)$

Now substituting this back into the

Expression for I:

$I = \sum_{n=0}^{n} ne^{n+1}e^{-n}\left(1-e^{-1}\right) = \left(1-e^{-1}\right)\sum_{n=0}^{9} ne^{1}$

The sum $\sum_{n=0}^{9} n$ is the sum of the first 9 natural numbers:

$\sum_{n=0}^{9} n = 0+1+2+....+9 = \frac{9(9+1)}{2} = 45$

Thus, we have:

$I = \left(1-e^{-1}\right)e.45$

This gives us:

$I = 45e\left(1-e^{-1}\right) = 45e-45$

The value of the integral I is:

$I = 45(e-1)$

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72. We have a function $F:\mathrm{R}\to\mathrm{R}$ that is thrice

Differentiable. We know:

$-F(1) = 0$

$-F(3) = -4$

$-F(x) < 0 \text{ for all } x \in (1,3)$

We also have a function defined as

$f(x) = xF(x).$

Given Integrals

We are given two integrals:

1. $\int_1^3 x^2F(x)dx = -12$

2. $\int_1^3 x^3F^n(x)dx = 40$

Finding $F(x)$

Since $F(x)$ is thrice differentiable, we can assume a quadratic form for $F(x)$:

$F(x) = ax^2 + bx + c$

We will use the conditions $F(1) = 0$ and

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$F(3)=-4$ to find a,b and c.

From $F(1)=0$:

$a(1)^2+b(1)+c=0\Rightarrow a+b+c=0 \quad (1)$

From $F(3)=-4$:

$a(3)^2+b(3)+c=-4\Rightarrow 9a+3b+c=-4 \quad (2)$

Differentiate $F(x)$

Now we differentiate $F(x)$:

–First derivative: $F'(x)=2ax+b$

–Second derivative: $F''(x)=2a$

Evaluate the Integrals

Using the first integral:

$$\int_1^3 x^2F'(x)\,dx=\int_1^3 x^2(2ax+b)dx=\int_1^3(2ax^3+bx^2)\,dx$$

Calculating this integral:

$$=2a\left(\frac{x^4}{4}\right)_1^3+b\left(\frac{x^3}{3}\right)_1^3$$

Calculating this integral:

$$=2a\left(\frac{81}{4}-\frac{1}{4}\right)+b(27-1)=2a.20+26b$$

$$=40a+26b=-12 \quad (3)$$

$$\int_1^3 x^3F''(x)\,dx=\int_1^3 x^3(2a)\,dx=2a\left(\frac{x^4}{4}\right)_1^3=2a$$

$.20=40\Rightarrow a=1 \quad (4)$

Substitute a into Equations

Substituting $a=1$ into equation (3):

$40(1)+26b=-12\Rightarrow 40+26b=-12\Rightarrow 26b=-52\Rightarrow b=-2$

Find c

Substituting $a=1$ and $b=-2$ into equation (1):

$1-2+c=0\Rightarrow c=1$

Final From of $F(x)$

Thus, we have:

$F(x)=x^2+2x+1=(x-1)^2$

Find $f(x)$

Now, we can find $f(x)$:

$f(x)=xF(x)=x(x-1)^2=x^3-2x^2+x$

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73. **Understanding the inequalities**

The inequality $|x^2-2| \le y$ can be split into two cases:

1. $y \ge x^2-2$
2. $y \le 2-x^2$

Thus, we have:

$x^2-2 \le y \le x$

And

$y \le 2-x^2$

Finding the intersection points

To find the area, we need to determine the points where these curves intersect. We set:

1. $y = x$
2. $y = x^2-2$

Setting $x = x^2-2$:

$x^2-x-2=0$

Factoring gives:

$(x-2)(x+1)=0$

Thus, $x=2$ and $x=-1$.

Next, we set:

1. $y = x$
2. $y = 2-x^2$

Setting $x = 2-x^2$:

$x^2+x-2=0$

Factoring gives:

$(x-1)(x+2)=0$

Thus, $x=1$ and $x=-2$

Finding the area

The area A can be computed by integrating the difference between the upper and lower curves from the leftmost intersection point to the rightmost intersection point.

The relevant intervals for integration are from $x=-2$ to $x=2$.

1. From $x=-2$ to $x=-1$:

$-$upper curve: $y = 2-x^2$

- –Lower curve: $y = x^2$-2

2. From $x=-1$ to $x=1$:

Uppercurve : $y = x$

$-Lowercurve : y = x^2-2$

3. From $x=1$ to $x=2$:

$-$Upper curve : $y = x$

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$-$Lower curve : $y = x$

Setting up the integrals

The area A is given by:

$$A = \int_{2}^{-1} \left(\left(2 - x^2\right) - \left(x^2 - 2\right) \right) dx + \int_{-1}^{1} \left(x - \left(x^2 - 2\right) \right) dx + \int_{1}^{2} \left(\left(2 - x^2\right) - x \right) dx$$

Evaluating the integrals

1. For $\int_{-2}^{-1} \left(4 - 2x^2\right) dx$:

$$= \left(4x - \frac{2}{3}x^3 \right)_{-2}^{-1} = \left(-4 + \frac{2}{3} \right) - \left(-8 + \frac{16}{3} \right) = -\frac{10}{3} + \frac{8}{3} = -\frac{2}{3}$$

2. for $\int_{-1}^{1} \left(x + 2 - x^2\right) dx$:

$$= \left(\frac{x^2}{2} + 2x - \frac{x^3}{3} \right)_{-1}^{1} = \left(\frac{1}{2} + 2 - \frac{1}{3} \right) - \left(\frac{1}{2} - 2 + \frac{1}{3} \right) = 4$$

3. For $\int_{1}^{2} \left(2 - x^2 - x\right) dx$:

$$= \left(2x - \frac{x^3}{3} - \frac{x^2}{2} \right)_{1}^{2} = \left(4 - \frac{8}{3} - 2 \right) - \left(2 - \frac{1}{3} - \frac{1}{2} \right) = 1$$

Summing the areas

Combining these areas gives:

$$A = \left(-\frac{2}{3} + 4 + 1 \right) = \frac{11}{3}$$

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74. **Understand the Condition for G.P.**

For three numbers $a, b,$ and c to be in G.P., the condition that must hold is:

$b^2 = ac$

This means that the middle term squared should be equal to the product of the other two terms.

Express the Numbers in Terms of Exponents

Let's denote the numbers in the set as

$3^1, 3^2, 3^3, ..., 3^{10}$. We can represent these numbers by their exponents:

$1, 2, 3, ..., 10$

If we select three numbers $3^a, 3^b,$ and 3^c such the $a < b < c$, then the condition for these to be in G.P. translates to:

$b - a = c - b$

This implies that b is the average of a and c, or equivalently, $a, b,$ and c must be in arithmetic progression (A.P.).

Identify Possible Values of $a, b,$ and c

Since b must be the average of a and c, we can express a and c in terms of b:

$a = b - d$ *and* $c = b + d$

Where d is a positive integer.

Determine the Range for b

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The values of a and c must remain within the bounds of the set, which means:

$1 \le a < b < c \le 10$

This gives us the following inequalities:

$1 \le b - d$ and $b + d \le 10$

From these inequalities, we can derive the limits for b and d.

Calculate the Possible Values of b

From $1 \le b - d$ we have :

$d \le b - 1$

From $b + d \le 10$ we have :

$d \le 10 - b$

Thus, combining these gives:

$d \le \min(b - 1, 10 - b)$

The maximum value of d can be calculated for each possible b.

Count the Valid Combinations

Now we can count how many valid pairs (a, c) can be formed for each b:

– For $b = 2$: d can be $1 (1 \text{ way})$

– For $b = 3$: d can be 1 or $2 (2 \text{ ways})$

– For $b = 4$: d can be $1, 2$ or $3 (3 \text{ ways})$

– For $b = 5$: d can be $1, 2, 3,$ or $4 (4 \text{ ways})$

– For $b = 6$: d can be $1, 2, 3,$ or $4 (4 \text{ ways})$

– For $b = 7$: d can be $1, 2,$ or $3 (3 \text{ ways})$

– For $b = 8$: d can be $1,$ or $2 (2 \text{ ways})$

– For $b = 9$: d can be $1 (1 \text{ way})$

Sum the Ways

Now, we sum the number of ways:

$1 + 2 + 3 + 4 + 4 + 3 + 2 + 1 = 20$

Calculate $N / 5$

Finally, since $N = 20$, we calculate:

$N / 5 = 20 / 5 = 4$

Final Anwer

The value of $N / 5$ is 4.

75. Equation of circle passing through $(6, -4), (22, 12)$ is $(x - 10)^2 + (y - 8)^2 = 160$

$(13 - k, 13 + k)$ lie on this circle

$\Rightarrow (3 - k)^2 + (5 + k)^2 = 160$

$2k^2 + 4k + 34 - 160 = 0$

$k^2 + 2k - 63 = 0$

$(k + 1)^2 = 64$

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$k+1=\pm 8$

$k=7,-9$

$(13-k,13+k)\equiv(6,20)\ or\ (22,4)$

$r^2 = 160$

$k = 7$

$k+r^2=167$

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